

File permissions in Linux

Project description

In this project I will be showcasing how to use the Linux commands. We can use Linux commands for various type of tasks in the CLI.

Check file and directory details

By using the `ls -la` command we can display all of the files inside a directory, including the hidden files. In the screenshot, we can see the `.projects_x.txt` is a hidden file.

```
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w--- 1 researcher2 research_team   46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

Change file permissions

To manage file system security, the `chmod` command is used to alter permissions. In this demonstration, I used `chmod o-w project_k.txt` to remove write access for 'others,' effectively preventing unauthorized modifications to the file.

```
researcher2@5d738f0f927b:~/projects$ chmod o-w project_k.txt
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w--- 1 researcher2 research_team   46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

Change file permissions on a hidden file

In response to `project_k.txt` being archived, its file permissions were updated to prevent any further changes. The command `chmod u-w,g-w,g+r` was used to remove write permissions for both the user and group, while ensuring the group retained read access.

```
researcher2@3213bbc1d047:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@3213bbc1d047:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 ..
-r--r----- 1 researcher2 research_team  46 Dec 20 15:36 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec 20 15:36 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Dec 20 15:36 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec 20 15:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec 20 15:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec 20 15:36 project_t.txt
researcher2@3213bbc1d047:~/projects$
```

Change directory permissions

The security requirement is to grant exclusive access of the `drafts` directory to the `researcher2` user. This is accomplished by removing the execute permission for any other user or group, which blocks them from accessing the directory.

Using the `chmod g-x drafts` we can remove the execution permission for the drafts folder.

```
researcher2@5d738f0f927b:~/projects$ chmod g-x drafts
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-r--r----- 1 researcher2 research_team  46 Dec  2 15:27 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

Summary

We can use various Linux commands to work with directories and files and their permissions. Permissions are important as a way of protecting the organization from malicious actors trying to gain access to files inside the organization. The organization is kept safe by only allowing certain users and groups a level of access only needed to accomplish their daily tasks.